

38.1 Variance functions

Fitting a generalized linear model needs a family, a link and a linear predictor. [Section 34.3](#) noticed that the family enters the likelihood equations

$$\sum_{i=1}^n x_{ij} \frac{w_i h'(\eta_i)}{V(\mu_i)} (y_i - \mu_i) = 0, \quad j = 1, \dots, p, \quad (38.1.1)$$

only through the variance function V : two densities with the same V produce the same estimate. The apparatus of Definition 34.1.1 — a natural parameter, a cumulant function, a density that integrates to one — *derives* [Eq. 38.1.1](#) and then disappears from it. This section puts the variance function first, and the model becomes a mean and a variance and nothing else.

38.1.1 Specifying the second moment only

Definition 38.1.1. A variance specification

Let $Y_1, \dots, Y_n$ be independent, let $w_1, \dots, w_n > 0$ be known weights, let g be a strictly monotone differentiable link with inverse h , and let V be a positive, continuously differentiable function on an open interval $\mathcal{M} \subseteq \mathbb{R}$. A **variance specification** is the pair of assumptions

$$E(Y_i) = \mu_i = h(\mathbf{x}_i^\top \boldsymbol{\beta}), \quad \text{Var}(Y_i) = \frac{\phi V(\mu_i)}{w_i}, \quad (38.1.2)$$

with $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\phi > 0$ unknown. The function V is the **variance function** and ϕ the **dispersion**. No distribution is assumed for Y_i beyond these two moments.

The definition is weaker than Definition 34.2.1 in saying nothing about higher moments, the support, or whether any distribution with these moments exists, and stronger in leaving ϕ free even for a V that came from a family in which it was fixed. The specifications in constant use are

name	$V(\mu)$	w_i	ϕ	$\mathcal{M}$
quasi-normal	1	1	free	$\mathbb{R}$
quasi-Poisson	μ	1	free	$(0, \infty)$
quasi-binomial	$\mu(1 - \mu)$	m_i	free	$(0, 1)$
quasi-gamma	μ^2	1	free	$(0, \infty)$
power	μ^ξ	1	free	$(0, \infty)$
NB2	$\mu + \mu^2/\kappa$	1	1	$(0, \infty)$

where y_i is a group proportion in the quasi-binomial row and m_i the number of trials behind it. A value $\phi > 1$ is **overdispersion** relative to the family that supplied V , and $\phi < 1$ **underdispersion**. The first four rows are variance functions of Definition 34.1.2 with the dispersion set free; the last is that of Definition 37.4.1(a), for which a genuine likelihood exists.

Remark (When is a variance specification a distribution?). For ϕ free and $V(\mu) = \mu^\xi$, Jørgensen's classification of the Tweedie families (Exercise (C2)) settles whether some exponential dispersion family has exactly these moments: one exists for $\xi \leq 0$ and for $\xi \geq 1$, and for no ξ strictly between. Existence is not usefulness. At $\xi = 1$ the Tweedie family with dispersion ϕ is ϕW with W Poisson of mean μ/ϕ : the right two moments, but on the multiples of ϕ rather than the integers. So no *exponential dispersion family* on the counts has mean μ and variance $\phi\mu$ for one ϕ and all μ . A family of counts with exactly those two moments does exist — the NB1 model of Definition 37.4.1(b) — but it is not an exponential dispersion family, and its likelihood equations are not [Eq. 38.1.1](#); that is the point of the third subsection of this section.

Warning. The quasi-binomial specification is empty when $m_i = 1$. An ungrouped binary response has $Y_i^2 = Y_i$, so $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$ exactly for every distribution on $\{0, 1\}$ with mean μ_i , and there is no room for a ϕ . A dispersion estimate far from 1 in an ungrouped logistic fit is a statement about the mean model — an omitted term, a wrong link, a few gross outliers — never about the variance, the second-moment counterpart of Theorem 34.5.5(c).

38.1.2 Two ways to inflate a binomial

Where a dispersion is available it can take more than one shape, and the shapes are distinguishable. Suppose observation i averages m_i dependent binary trials.

Lemma 38.1.1. Exchangeable trials

Let $Z_1, \dots, Z_m$ be binary variables with common mean π and $\text{Cov}(Z_s, Z_t) = \rho\pi(1 - \pi)$ for all $s \neq t$, so that ρ is their common correlation, and let $\bar{Z} = m^{-1} \sum_t Z_t$. Then

$$\mathbb{E}(\bar{Z}) = \pi, \quad \text{Var}(\bar{Z}) = \{1 + \rho(m - 1)\} \frac{\pi(1 - \pi)}{m}, \quad (38.1.3)$$

and $\rho \geq -1/(m - 1)$, with $\rho \geq 0$ whenever the trials are conditionally independent and identically distributed given some unobserved variable.

Proof. Each Z_t has variance $\pi(1 - \pi)$, so the stated covariance is ρ times it and $\text{Var}(\bar{Z}) = m^{-2}\{m\pi(1 - \pi) + m(m - 1)\rho\pi(1 - \pi)\}$, which is [Eq. 38.1.3](#); nonnegativity gives $1 + \rho(m - 1) \geq 0$. If the Z_t are independent given U with $\Pr(Z_t = 1 \mid U) = \Pi$ the same for each t , the law of total covariance gives $\text{Cov}(Z_s, Z_t) = \text{Var}(\Pi) \geq 0$. ■

The variance specification suggested by [Eq. 38.1.3](#) is

$$V_\rho(\mu) = \{1 + \rho(m_i - 1)\} \mu(1 - \mu), \quad w_i = m_i, \quad \phi = 1,$$

in which the inflation grows with the group size, against the quasi-binomial $V(\mu) = \mu(1 - \mu)$ with $w_i = m_i$ and a constant ϕ . The two agree at one group size and nowhere else ([Figure 38.1.1](#)): at $\phi = 1.8$ and $\rho = 0.05$ they cross at $m = 17.0$ and differ by a factor 1.639 at $m = 40$.

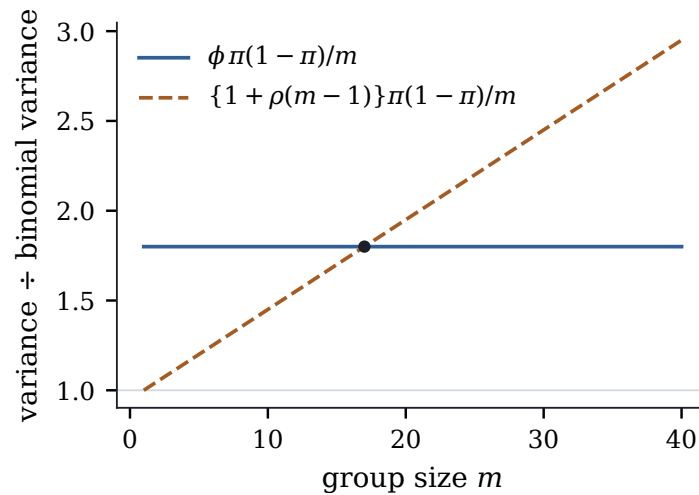


Figure 38.1.1: Two variance functions for a proportion from m trials, as multiples of $\pi(1-\pi)/m$. A constant dispersion inflates every group equally; exchangeable trials inflate large groups more, by Eq. 38.1.3.

Spread-out group sizes separate them, by a plot of squared Pearson residuals against m_i ; equal ones do not.

```
import numpy as np

m_grid = np.arange(1, 41)
phi, rho = 1.8, 0.05
inflated = np.full_like(m_grid, phi, dtype=float)      # V(pi) = phi pi(1-pi)/m
exchangeable = 1 + rho * (m_grid - 1)                  # V(pi) = {1+rho(m-1)} pi(1-
    pi)/m
crossing = 1 + (phi - 1) / rho                          # the one group size where
    they agree
print(f"the two variance functions agree at m = {crossing:.1f}")
print(f"at m = 40 they differ by a factor {exchangeable[-1] / phi:.3f}")
```

[Run this code online](#)

The second specification is not an invention of the quasi-likelihood literature: the **beta-binomial** distribution, a binomial mixed over a beta success probability, has exactly this variance (Exercise (B1)), so a mixture and a correlation leave the same trace in the second moment.

38.1.3 A variance function does not determine a fit

A variance function settles the estimating equations, which is less than settling the fit: two models can share a variance function and disagree about $\hat{\beta}$, because the second model's likelihood equations need not be Eq. 38.1.1. The NB1 model of Definition 37.4.1(b) has variance $\phi\mu_i$ with $\phi = 1 + \alpha^{-1}$, the quasi-Poisson variance function exactly; but its score Eq. 37.4.6 does not contain the residual $y_i - \mu_i$ at all, and the two estimates differ in general (Exercise (B3)), as Section 38.4 shows in numbers. A *model* is a set of distributions; a *variance specification* is two moment assumptions plus the rule Eq. 38.1.1 for turning them into an estimate.

38.1.4 What the missing density costs

Dropping the density is not free. There is no maximized likelihood, so Proposition 34.5.3 loses one of its three members and [Section 38.3](#) must supply substitutes; no Akaike or Bayesian criterion, since both are $-2\ell(\hat{\beta})$ plus a penalty (Definition 29.2.2), though cross-validation (Definition 29.3.1) survives; and no predictive distribution, so a fitted quasi-Poisson model cannot say what fraction of units will record zero, simulate a data set, or be checked by a predictive comparison, where a negative binomial or beta-binomial model does all three. What survives is the estimate, its consistency, its asymptotic normality, and a covariance estimate that does not need the variance function to be right — the subject of the next two sections.

38.1.5 Exercises

38.1.5.1 A. Check your understanding

Exercise. A1

For each of the following, say whether a dispersion parameter $\phi \neq 1$ is meaningful, and why: (a) counts of insurance claims per policy; (b) the proportion of a litter of 12 pups showing a defect; (c) an indicator of whether a patient survived; (d) the proportion of 500 sampled voters supporting a candidate, the voters sampled in clusters of 10 households.

Solution.

- (a) Yes: nothing ties a count's variance to its mean, and heterogeneity between policies inflates it (Lemma 37.4.1(a)). (b) Yes, with $w_i = 12$: correlation between littermates, or variation between litters, inflates the variance by [Lemma 38.1.1](#). (c) No: a binary response has variance $\mu(1 - \mu)$ whatever else is true. (d) Yes, but the group size is the cluster, not 500; using $m = 500$ in [Eq. 38.1.3](#) would overstate the inflation.

Exercise. A2

Two analysts fit grouped binomial data with group sizes between 5 and 60, one using a constant dispersion ϕ and one using the exchangeable form [Eq. 38.1.3](#) with correlation ρ . They report $\hat{\phi} = 2.5$ and $\hat{\rho} = 0.04$. At which group size do the two fitted variances agree? For which groups does the first analyst's model claim the larger variance?

Solution. Setting $1 + \rho(m - 1) = \phi$ gives $m = 1 + (\phi - 1)/\rho = 1 + 1.5/0.04 = 38.5$; below it the constant-dispersion model claims the larger variance, above it the smaller.

38.1.5.2 B. Practice

Exercise. B1

Let Π have a beta distribution with parameters $a, b > 0$, so that $E \Pi = \mu = a/(a + b)$ and $\text{Var } \Pi = \mu(1 - \mu)/(a + b + 1)$, and let $m\bar{Y}$ be binomial (m, Π) given Π . Show that $E \bar{Y} = \mu$

and

$$\text{Var}(\bar{Y}) = \{1 + \rho(m-1)\} \frac{\mu(1-\mu)}{m}, \quad \rho = \frac{1}{a+b+1},$$

so the beta-binomial has the variance function of [Eq. 38.1.3](#). Deduce that the beta-binomial and an exchangeable-trials model with the same ρ cannot be told apart by first and second moments.

Solution. The conditional mean is Π and the conditional variance $\Pi(1-\Pi)/m$, so $E\bar{Y} = \mu$ and, by the law of total variance and $E\Pi^2 = \text{Var}\Pi + \mu^2$,

$$\text{Var}(\bar{Y}) = \frac{E\{\Pi(1-\Pi)\}}{m} + \text{Var}\Pi = \frac{\mu(1-\mu) - \text{Var}\Pi}{m} + \text{Var}\Pi = \frac{\mu(1-\mu)}{m} \{1 + (m-1)\rho\}$$

with $\rho = \text{Var}\Pi / \{\mu(1-\mu)\} = 1/(a+b+1)$. The two specifications therefore give identical first and second moments for every m .

Exercise. B2

Under the power specification $\text{Var}(Y) = \phi\mu^\xi$, find the coefficient of variation $\sqrt{\text{Var}(Y)}/E(Y)$ as a function of μ . For which ξ is it constant? Which of the specifications in the table of this section have the property that a multiplicative change of units, $Y \mapsto cY$, leaves the specification in the same family with a new ϕ ?

38.1.5.3 C. Going deeper

Exercise. C1

Let $Y \mid \Lambda$ have mean Λ and variance $a\Lambda^\xi$, and let Λ have mean μ and finite variance τ^2 . Show that the marginal variance is $a E(\Lambda^\xi) + \tau^2$, equal to $a\mu + \tau^2$ when $\xi = 1$. For $\xi = 2$, use Jensen's inequality to show that it is at least $a\mu^2 + \tau^2$, so a mixture of quasi-gamma responses is overdispersed relative to the quasi-gamma specification with the same a .